ActInf GuestStream #016.2 ~ Mark Solms "Consciousness as Precision Op

GuestStream | Mark Solms | 2:04:33

Hello and welcome everyone. This is ActInfLab Guest Stream number 16.2. It is June 11th,

2022. We're here with Mark Solms. This is the second part in our discussion. We had an awesome

part one. Today we are going to have a recap on some of the points that were addressed in

the first part in 16.1 and then we're going to head into some unexplored territory. So Mark,

thanks so much for joining for these sessions and please take it away.

Thanks Daniel. Glad to be here again. I was just testing out sharing my screen a minute ago and it

seems as if I'm incapable of showing you my face and sharing my screen at the same time.

So take a good look at me. You're not going to see me for a while. Here comes my screen. Please guard.

There we are. Great. And here comes my pointer.

There we are. And here's my title.

The emphasis is on physiological and philosophical considerations rather than computational ones.

So I'll start with a brief recap. I made a few claims and I didn't get to the end of this list.

So I will just remind you what those claims were and then I will pause when I get to the point that we stopped at last time,

which was here, that affect is an extended form of homeostasis.

I started with the claim, which really is the main claim of my presentation.

It is that affect feeling is the foundational form of consciousness and that it is intrinsically conscious.

I quoted Freud, who more than anyone else introduced the notion into mental science,

that mental processes are not intrinsically conscious,

that much of our cognitive processing goes on unconsciously.

even he made this point, the point that I'm making,

which in his words was that it is surely of the essence of an emotion that we should be aware of it.

That is, that it should become known to consciousness.

Thus, the possibility of the attribute of unconsciousness would be completely excluded as emotions, feelings and affects are concerned.

And I emphasize the word feelings because there's a whole lot of reasons why some people claim that there are such things as unconscious emotions and unconscious affects.

So just to be absolutely clear what I'm talking about here, I'm talking about feeling and you can't have an unconscious feeling.

In other words, you can't have a feeling that you don't feel.

So whatever might be meant by unconscious emotions and unconscious affects, they can't apply to feelings.

They can't be unconscious feelings.

But that might sound like a semantic point, just a matter of words, that you can't have a feeling that you don't feel.

That's an oxymoron.

So I then went on to set out various reasons, empirical ones, why the claim that feelings are intrinsically conscious also has a solid empirical basis.

I drew attention to this discovery that was made in 1949 already, that the brain mechanisms for consciousness itself,

in other words, what wakes us up in the morning and puts us to sleep at night,

the sort of switching on and off of the lights of consciousness,

that this is the business of the reticular activating system.

The reticular activating system is the brain system for arousing consciousness,

for activating cortical processes and rendering them conscious.

I also mentioned the periaqueductal gray, which plays a crucial part in all of this,

together with the reticular activating system.

But I'll come back to that later.

For now, let's just focus on the reticular activating system.

To give you a sense of what I mean by how these structures,

their intrinsic function is the switching on of the lights of consciousness,

I pointed out that you can make a lesion or you can suffer a lesion as small as two cubic millimeters in size

in the parabrachial complex of the reticular activating system.

In other words, a lesion the size of a match head,

and that is enough in human beings reliably to obliterate consciousness entirely.

So I was saying, let's look at these structures if we're wanting to understand something of the intrinsic consciousness generating mechanisms of the brain.

And I made the point that if the cortex, which is where we normally shine our light in terms of looking for the neural correlates of consciousness,

if the cortex were the seat of consciousness, then there's an easily testable prediction,

which is that if you have a case in which there is no cortex, then the patient should not be conscious.

Not so.

Therefore, I showed you one such case.

This is one representative example of its type.

That here is a patient born with no cortex.

The condition is called hydranencephaly.

Here you see in an MRI scan of her brain that where cortex should be,

there is just cerebrospinal fluid.

And you see that her brainstem is perfectly intact.

So on the view that this brainstem area is where the consciousness is generated, that it doesn't require cortex.

This is a critical case.

On the brainstem view, she should be conscious.

On the cortical view, she should not.

And here she is.

And she's conscious.

As you can see, she wakes up in the morning.

She goes to sleep at night.

And in this sense, the lights are on.

But much more interestingly, she's not merely blankly awake.

Her wakefulness has a content and a quality.

And the quality that I'm talking about is affective quality.

You see how she responds to her baby brother being placed on her lap.

She responds with some form of pleasure.

And so there's a content to this mental state and a quality to this mental state that she's displaying.

Why that's important is because when we first learned in 1949 that the reticular activating system is prerequisite for all consciousness,

that there's no such thing as cortical consciousness without brainstem arousal of the cortex.

We had the view that the cortex provides the qualities and the contents of consciousness.

And the reticular activating system merely provides the quantitative dimension, the sort of level of consciousness.

So it's as if this was a power supply like a television set needs to be plugged in at the wall.

The reticular activating system is the prerequisite sort of booting up of the system.

But the television set itself, the cortex, where the contents and qualities of consciousness are processed,

could still be claimed to be the seat of consciousness and the reticular activating system merely as a power source.

This is why it was important to point out that this child is not only conscious in some blank sense of wakefulness without content and quality,

but rather that she displays affects.

She's emotionally responsive to her baby brother being placed on her lap,

just as this child, who also has no cortex, is emotionally responsive.

Here you can see she's responding with pleasure to a stimulus.

And this is quite generally the case for these children.

Here's Bjorn Merke's summary of his observations in many, many, many such children.

They express pleasure, they smile, they laugh, they show aversion and fussing by arching their backs and crying.

Their faces are animated by these emotional states.

They build up play sequences, they smile, they giggle, they laugh, they show great excitement, etc., etc., etc.

So I've highlighted all of those words to show that these two cases that I just showed you with no cortex,

they are conscious and they are responsive.

And in particular, they are emotionally responsive.

And on this, I base the claim that these children do have a quality,

that they do have a consciousness and this consciousness does have a quality and it does have content.

Many people are perplexed as to how this could happen, since they have no cortex.

How can they respond to things like their baby brother's been placed in their laps?

And so I just inserted this slide, I don't think I showed it last time,

just to point out that our sensory end organs, here's the example is the eye.

Same applies to the skin and to hearing and taste as well.

The optic nerve projects to the lateral geniculate and from there to the visual cortex,

but not only to the visual cortex, it also projects subcortically to the superior colliculi of the midbrain,

of the brain stem, which is immediately adjacent to the periaquiductal gray,

which I said earlier I was going to mention again.

So these children receive information in the brain stem, which is not conscious,

it's not cognitively conscious perception.

Conscious perception is generated in the cortex, but unconscious sensory information goes into the brain stem,

where it is responded to consciously by the emotional structures that are the main focus of what I'm talking to you about.

Now, of course, many of you, like many of my colleagues, will reasonably say,

well, how do you know that there's something it's like to be those children?

These might just be reflexes or instincts.

These might be the equivalent, these kids, of philosophical zombies.

In other words, they look as if they're conscious, they behave as if they're conscious,

but we can't know that they're actually conscious.

And so last time I tried to address this objection by drawing attention to what happens in cases who lose great swathes of their cortex

and because they haven't lost all of their cortex, they are able to describe to us what it's like to be them.

This is one way of getting around the objection that because these kids have no cortex and therefore can't declare their conscious states,

we can't be sure that they have conscious states.

So I focused, first of all, on a case who has a massive lesion of the prefrontal cortex.

The reason I did that is because this is a favorite part of the cortex for those who claim the cortex is the seat of consciousness,

like the global workspace theorists.

They say it all comes together in the prefrontal lobes.

All of this information that's processed in the posterior cortices is re-represented or accessed in the global workspace.

And this is where the sentient being, the subject of the mind, comes about.

And I pointed out last time that if that were the case, then again we have a falsifiable prediction.

A patient who has no prefrontal cortex, like this patient of mine, patient W, he has no prefrontal cortex,

but he has a sliver of language cortex.

He, if his sentient being was contingent upon the integrity of prefrontal cortex, he should not have sentient being.

And so I showed you what it's like.

I talked to him about what it's like to be him.

He claims to be consciously aware of his thoughts.

I asked him to imagine something for me, to imagine two dogs and a chicken, to see them in his mind's eye.

And then I asked him to count the legs.

I thought this would be a reasonable test of whether he's actually got conscious mental imagery.

And please note, the person I'm talking to refers to himself as I.

So he seems to think that he exists as a sentient being.

And I asked him how many legs there are in total, if you have two dogs and one chicken.

And to my disappointment, he said eight.

And when I questioned his answer, he pointed out that in his mind's eye, the dogs had ate the chicken.

So I thought that was maybe not a great joke, but it certainly suggested that there was somebody at home.

And I made the point last time that these patients generally, patients with massive frontal lesions,

that this tendency to make puerile jokes is considered to be a rather common part of the frontal lobe syndrome or the frontal lobe personality.

And this is part of a bigger story that I mentioned last time, which is that these patients are generally quite emotionally disinhibited.

And why that's important is because, remember, what we are considering here is the question as to whether the feelings that you saw in those patients,

those kids with no cortex, whether those feelings could be generated in the brain stem.

And the claim of corticocentric theorists like Joe Ledoux, for example, is that the feelings literally come about

when they are registered or re-represented or labeled or even named, some people claim,

that it's only once you are able to re-represent these subcortical survival circuits in declarative consciousness in prefrontal lobes,

that this is literally what brings the feeling about.

And I think it's quite interesting that these patients who have no prefrontal cortex don't have a dearth of feeling,

which is what you would expect if the machinery that brings feelings about is absent,

then they shouldn't be able to have feelings.

But in fact, what is generally accepted is these patients have an excess of feelings.

There's disinhibited emotionality in these patients.

I made much the same argument about the other major cortical area that is associated with the sentient self,

and that is the insular cortex.

Of course, this is associated above all with the work of Bud Craig,

but has been very widely accepted, that the feeling self comes about when the state of one's interceptive body

is re-represented in insular cortex.

And so, again, we have a falsifiable prediction.

If you take a case who has no insular cortex, like this patient of Damasio's, then there should be no sentient self present.

And I showed you this interview where Damasio speaks to him about his sense of self,

and the patient is perfectly adamant that he exists as a self.

He speaks about himself as I, I, you know.

And this interview ends with Damasio saying, you're aware that I'm aware,

and the patient B says, I'm aware that you're aware that I'm aware.

You know, and this patient, just like my patient W, is not deficient in emotionality.

In fact, all the basic emotions are present, including both bodily affects and emotional affects

and sensory affects.

And not only that, he's also a little too emotional.

And this is a little disinhibited in his emotionality.

And this is what we quite generally see with patients with insular lesion.

So, again, it's very hard to sustain the argument that the self actually comes into being in the cortex.

Now, of course, those two cases, those examples of their kinds,

patients with massive frontal cortical lesions, patients with massive insular lesions,

of course, they are just examples.

But the point is, the point I'm making now is that there's lots of cortex left in those cases.

So, the argument is, well, you know, maybe the sentient subject is generated in the remainder of their cortex.

And it's a bit of a circular argument, because remember, you know, the children who have no cortex,

we're told, well, how do we know that they're conscious, they can't report their conscious states.

I then gave these two outstanding examples, looking at the areas of cortex that are most bound up with sentient subjectivity,

according to cortical theories of consciousness.

And these patients seem to have intact sentient subjectivity.

But now I'm told, well, you know, the remainder of their cortex might be what's generating the consciousness.

So, we can't use only lesion methods.

There's no way out of that, on pass.

And so, I then showed you last time evidence of a different kind.

I showed you what happens if you stimulate reticular activating system nuclei.

There, again, there's an easily tested prediction.

The prediction is that if these brainstem nuclei, the reticular activating system and periaqueductal gray,

as I'm claiming, if these are the structures that generate consciousness and feeling,

then stimulation of these structures should stimulate conscious feelings.

And I showed you that here's a case who had a deep brain stimulator placed into the substantia nigra,

this part of her reticular activating system.

And within five seconds, it generated, it produced a profound depression.

The patient was actually suicidal.

She didn't want to live anymore.

This is a patient with no psychiatric history.

The electrode was placed in her brainstem for the treatment of Parkinson's disease and stimulated the wrong nucleus.

And that's how this came about.

Ninety seconds after the stimulator was switched off, the depressed feeling disappeared.

And the patient generously agreed to allow for further stimulation in the reticular activating system and out of it.

And it was only when that particular nucleus was stimulated that she fell into the depression.

So this is the kind of evidence.

And remember, again, I'm just giving you examples.

You can stimulate intense affective states by stimulating reticular activating nuclei and periaqueductal gray.

You get the greatest intensity and the greatest variety of affects from stimulating there.

And you get nothing of the kind from stimulating cortex.

So this is a different line of evidence suggesting that feelings are actually generated in the upper brainstem.

And then I showed you another line of evidence that was positron emission tomography of people in intense affective states.

Here you see research participants in states of sadness, here of anger, here of happiness, here of fear.

And in all instances, the activation is in the brainstem and the circuits arising from it.

The subcortical circuits arising from it.

That's what we see.

And the cortex is, by contrast, largely deactivated.

So this is a further line of evidence that the, you know, an entirely different line of evidence.

Remember, we've got lesion evidence.

Then we've got deep brain stimulation evidence.

Here we've got positron emission imaging evidence.

That the part of the brain that's generating the feelings is the part of the brain that switches on the lights.

And this is why I'm claiming that the basic, the foundational form of consciousness, this prerequisite form is affect.

Affect is the elemental form of consciousness.

A further line of evidence is pharmacological manipulations.

If you tinker with the neuromodulators that are sourced in these reticular activating nuclei, like, for example, noradrenaline or serotonin or dopamine, all of which are sourced in the reticular activating system.

Serotonin is, of course, regularly pharmacologically manipulated for the treatment of depression.

Dopamine for the treatment of psychosis.

Noradrenaline for the treatment of anxiety.

Noradrenaline is sourced in locus-cerulius complex.

Serotonin in the RAFE nuclei.

Dopamine, at least the one that's important for psychosis in the ventral tegmental area.

All of these are parts of the reticular activating system.

If all that this system did was switch on blank wakefulness, it might be of interest to anesthetists.

But, in fact, it is the main target of the drugs of psychiatrists who are treating emotional disorders by manipulating the chemistries that are the source nuclei for which are in the reticular activating system.

So, on the basis of all of that, I argued, remember, that affect is the foundational form of consciousness and it's intrinsically conscious.

I said that feelings, not only on semantic grounds, I'm saying that feeling is the basic form of consciousness.

The reason I'm saying that is because the basic consciousness generating tissues of the brain, the reticular activating system, which is prerequisite for the activation of consciousness in cortex, that these structures generate feeling.

That feeling, therefore, is prerequisite, is foundational for all forms of consciousness.

I then made, this is a sort of a sidebar.

The reason I went into this third point is that, namely, that the claim that affect is not synonymous with inter-receptive inference.

In other words, that it's not just an inter-receptive form of perception, as opposed to the extra-receptive forms, which have been the major focus of consciousness studies over the last few decades.

In other words, vision being the main focus, I'm saying that affect is not, and why I say this is because this is increasingly being argued.

In fact, it was argued by Bud Craig himself that affect is just inter-receptive perception.

The perception of the state of the state of the state of the outside world in the more typical form or extraceptive form of perception that's been equated with consciousness.

I made the point that there's good reason to believe that affect isn't just a sixth modality of perception.

And the evidence that I presented for that was of various kinds.

For example, that there are affects which are clearly not inter-receptive.

For example, getting a fright or being startled or feeling pain when a pin is stuck into your finger.

These are all extra-receptive forms of stimulation, and yet they arouse affective responses.

So affect clearly is not uniquely inter-receptive.

And then I also made the point that there are many inter-receptive perceptual states which are not affective.

You can feel your tummy grumbling, or you can feel your heart beating, or you can feel your lungs expanding.

These are not intrinsically affective phenomena.

So inter-receptive perception can happen without affect, and affect can happen without inter-receptive perception.

So I was trying to draw a line under the idea that affect is just another modality of perception.

I think that by detaching affect from perception casts some new light on the heart problem.

And so that's where I went next.

And I reminded you that David Chalmers, who coined the heart problem, said that when we look at functions, perceptual functions like vision,

which was, as I said, the model example derived from the, or grounded upon the assumption that consciousness is a cortical phenomenon,

cortical vision became the model example of consciousness.

This was following Crick's initiative in the mid-1990s.

Crick's idea was that if we can identify the neural correlate of consciousness in the case of cortical vision,

then we can generalize from that, by discerning the mechanism whereby visual information processing gets turned into conscious vision in the cortex,

by isolating that mechanism, we will be able to understand the nature and function of consciousness.

And Chalmers said that that's not true.

If you isolate, identify the mechanism of visual information processing like this map here does,

it doesn't tell you anything about why there's something it is like to see.

And he used the well-known knowledge argument of Frank Jackson, the story about Mary, the visual neuroscientist,

who knows everything about all of this.

And I slightly simplify the story by saying, well, let's imagine that Mary is blind,

even though she knows everything about the functions of cortical vision,

she knows everything about the mechanism whereby visual information is processed in the cortex.

Because she's blind, she knows nothing about what it is like to see.

And if she were to be gifted suddenly with a normal sight,

then she would learn something completely new about vision.

She would learn what it is like to see red and blue,

what blueness and redness, et cetera, are like,

none of which is accounted for by this information processing flow diagram.

None of what she knew about the functional mechanisms of cortical vision

would have prepared her for what it is like to see.

In other words, there's something else about visual information processing

other than the sort of things that mechanistic functionalist dissections like this provide us with.

This mechanistic account doesn't predict that there should be anything that it is like to see.

And this is the grounds upon which people like Chalmers say that an account of the functional mechanism

of cognitive and perceptual processes doesn't tell us anything about why there's something it is like to perform these processes.

And this is the essence of the hard problem.

I said that I thought that this might be because they were looking in the wrong place,

that visual perception and all forms of cortical perception,

and indeed not only perception but learning and cognition more generally,

that this can readily go on unconsciously,

that these are not intrinsically conscious processes,

that it is perfectly possible to see and to recognize faces

and to read with comprehension and even to discriminate colors.

These are all uniquely cortical processes,

and they can all go on in the dark, as it were.

In other words, you do not have to be conscious of what you're perceiving in order to perceive it.
And so this casts a rather different light on Chalmers' point
that all of this information processing can go on in the dark.
So consciousness isn't accounted for by our normal functionalist mechanistic way of doing cognitive neuroscience.
So I was saying, well, that's because they're looking at functions like perception
and functions like learning, which are not intrinsically conscious processes.
I drew your attention to this other statement of Chalmers' from his famous paper,
in which he says, in summary,
there is no cognitive function such that we can say in advance
that the explanation of the function will automatically explain experience.
And my point was, well, that's because they're talking about cognitive functions.
Could Chalmers make the same statement if we were talking about affective functions?
I'm saying, no, he could not.
But I'm saying that we can say in advance that explanation of the function of feeling
will automatically explain experience.
Because the function of feeling is to feel.
It's intrinsically conscious, is this function.
Unlike vision and perception in general and cognition as a whole,
none of that is intrinsically conscious.
But feeling, affective feeling, is intrinsically conscious.
You could not understand the mechanism of affective feeling
if it didn't account for why it feels like something,
because that's the whole bloody point of feeling.
So this question of Chalmers' why is the performance of these functions
accompanied by experience?
Why doesn't all this information processing go on in the dark,
free of any inner feel?
I'm saying that that question is perfectly reasonable
when asked of these cognitive functions,
which are not intrinsically conscious.
That kind of information processing can go on in the dark,
free of any inner feel.
But that is not true of feeling, of affective feeling.
And so my claim, and this is where we got to last time,
this is now, I'm now starting with new arguments.
My claim is that if we can identify the functional mechanism of affect,
of affective feeling, then we will be able to explain

why this sort of information processing doesn't go on in the dark.
So I hope that that's clear.
That's kind of like my main point.
My main point is we've been looking in the wrong place.
We've been looking to cortical vision and cortical perceptual
and cognitive processes in general,
in order to isolate the neural correlative consciousness.
And Chalmers has said, well, it doesn't work.
You can isolate the mechanism whereby this sort of,
these sorts of information processing goes on,
and it doesn't tell you anything about why there's something
it is like to see, et cetera.
And so my argument is, yes, that's true of those processes,
but it's not true of affect.
Affect, remember what I've said to you in my summary now today,
that this is the most concentrated consciousness-generating tissue
that there is.
The reticular activating system and periaqueductal gray
is where the lights are switched on.
And that you need, you can switch those lights off
with as small a lesion as two cubic millimeters in extent.
So this is the place where we should obviously have been looking
in the first place in order to understand,
to identify the neural correlative consciousness
and to understand its mechanism.
And much more important than that is the fact that
the kind of consciousness it generates, namely feeling,
is, it is an intrinsically conscious mechanism.
That feeling, it wouldn't exist if it wasn't felt.
And so unlike vision and learning and cognition in general,
which does not have to be felt.
Affect does have to be felt.
That's the whole point of affect.
So I'm saying that if we could understand the mechanism of affect,
in other words, the mechanism whereby feeling comes about,
then we might make some progress
with this hard problem of consciousness.
So remember, that's how far we got last time.

And now I go on to, and I'm now going to slow down a little bit,
because I'm now going to argue,
try to identify what the functional mechanism of affect is.
But I thought perhaps before I do that,
I should pause for a moment in case there are any questions
arising from my summary of my argument so far.
I'll carry on.
Thank you, Mark.
Okay.
Thanks.
Thanks.
So you see, I'm saying affect is an extended form of homeostasis.
And by the way, this is not my argument.
This is an argument that was first, to my knowledge,
first introduced by Jark Panksepp in the 1990s,
subsequently popularized by Antonio Damasio.
So I'm just summarizing my version of what is, in fact,
now a good 25-year-old argument.
Homeostasis is probably the most basic biological mechanism.
It could be said that homeostasis is what enables living organisms
to resist the second law of thermodynamics.
In other words, it is what enables them to be living organisms.
They don't just dissipate.
They remain in an organized form.
And the basis of this is homeostasis.
That rather than just explore all possible states,
we living things have to remain within highly specific states.
And these are called the settling point or the set point,
the viable ranges of the organism.
And this applies across multiple different parameters.
Let me use the example of core body temperature.
For those of us who think in degrees Celsius,
you know, we have to remain between 36.5 and 37.5 degrees Celsius.
That's where we need to be.
If we deviate too far from that very narrow range,
then we are at risk of dying.
And that doesn't apply only to temperature.
It applies to water, to oxygen, to salt, to sugar,

to blood pressure, to all sorts of things about our bodies.
They have to remain within very narrow ranges
that are viable with the preservation of our living state.
If we move outside of those ranges,
in other words, if we explore all possible states,
then we die.
So we have to work against that dissipative trend.
As I said, that is this entropic trend.
We have to resist entropy and remain within our viable states.
Very narrow, specific ranges of viability
across these multiple dimensions of our physiology.
And this is why I say that homeostasis is what keeps us alive.
Now, a deviation from that settling point then becomes a demand for work.
The organism has to do something to return itself to its viable bounds.
And that's the basic mechanism of homeostasis.
That's how all of those autonomic functions I was talking about earlier,
like core body temperature and blood gas balance and so on,
that's how they work.
If you're moving outside of your range,
you have to do something to return yourself into the range.
That is what homeostasis is.
It's the mechanism that returns us back to our viable range.
Now, what I'm saying is that affect is an extended form of homeostasis.
And how it extends homeostasis is that when we move out of our viable range,
we feel it.
We feel an unpleasant quality.
That means I am moving out of my viable range.
And by contrast, if you're moving back towards your viable range,
you have pleasurable feeling.
This seems to be the basic function of affect.
It enables the organism to know how it's doing
in terms of its organismic viability.
So the organism feels when it's moving outside of its viable range as unpleasure,
which means this is bad.
And it feels moving back towards its viable range,
back towards its ideal settling point as pleasure,
which means this is good.
This predicts my survival.

This predicts my demise.
feelings are rooted in a value system.
In other words, that there is something good and there is something bad.
And what is good is to survive, and as it happens to reproduce.
And what is bad is to not do so.
This is, of course, the basic value system that underwrites all of life.
This is the value system that underwrites natural selection.
So feelings are rooted in that value system.
And what they do is they enable the organism subjectively,
the organism itself, to know whether it's moving out of its viable bounds.
In other words, whether it's doing something bad within that value system
or something good within that value system.
That's what feelings do.
They enable the organism to know how they are doing within that value system.
Now, why does this get added to homeostasis?
Because not all homeostasis is felt.
Most homeostasis is entirely autonomic.
I was mentioning earlier blood pressure, for example.
When you move out of your viable range,
your heart rate changes and your blood vessels dilate,
in order to return you to your viable blood pressure.
You don't need to know.
In fact, you do not know anything about it.
In fact, it's clinically notorious how blood pressure regulation works,
because you can be way out of your viable bounds and know nothing about it.
And I'm just using blood pressure as an example.
There are many, many, many ways in which your autonomic nervous system
maintains you within your viable bounds without you knowing anything about it.
So why do we need to feel it?
Well, what feeling adds is when you're in a situation of uncertainty,
a situation where you do not have a readily pre-prepared reflex,
which returns you to your viable bounds.
So you don't need to feel how you're doing if you don't have to make any choices.
If you have automatic predictions, which return you by reflex,
like in the case of blood pressure regulation, to your viable bounds,
then you, the sentient being, have no part to play in the process.
Where feeling comes into its own is where the organism finds itself in a state of uncertainty.
For example, in a novel situation for which its innate pre-wired, as it were, reflexes have no preparedness.

You find yourself in a state of surprise.

And now what you're going to do, if there's no reflexive solution available,

then feeling enables the organism to make choices.

Choices have to be rooted in a value system.

There has to be a good choice and a bad choice.

Otherwise, it's random.

So whether you're doing the right thing or the wrong thing is announced to you by how it feels.

So this enables voluntary action.

I really must emphasize that.

This enables you to choose.

It enables you to decide for yourself what to do.

Things are getting worse.

So I'm going to change my mind.

Things are getting better.

I will stick with this policy.

This is working.

So these are not hardwired.

These are not innate predictions.

These are choices made on the fly by the organism here and now.

And they enable the organism to survive in states of uncertainty.

In other words, in unpredicted situations.

In other words, in novel situations.

And God knows there are many of those in life.

So just to put flesh to those bones, let me give you an illustrative example.

Normally, respiratory control is autonomic.

You don't need to make any decisions about breathing.

It just happens automatically.

So you automatically remain within your viable bounds in terms of the ratio of oxygen to carbon dioxide.

But that's only when the normal autonomic reflexes manage the situation.

Because you are in predictable circumstances.

Now, imagine that you are in a carbon dioxide filled room.

Now, suddenly you move out of your viable bounds.

And breathing normally, normal reflexive regulation of your respiratory system doesn't work.

Because ordinary breathing in this carbon dioxide filled room is going to rapidly lead to your demise.

So now you've got to do something.

You've never been in a burning building before.

Let alone this particular one.

So you have no reflex or instinct.

No innate prediction as to what to do.

So you feel your way through the situation.

And please note at this point, at the point when you find yourself in this unexpected situation,

this is where your need for oxygen becomes conscious.

So this is a very important point.

An otherwise autonomic function now becomes a conscious function.

You feel what we call air hunger or suffocation alarm.

And you now move about in this building, deciding which way to go.

Remember, you have no prior knowledge of what to do.

And it's only on the basis of how it feels.

So if, for example, you go upstairs and the oxygen supply diminishes, you feel worse air hunger.

If you go downstairs and there there's a greater provision of oxygen, then you feel better.

You feel relief from the suffocation alarm.

And so the feelings tell you whether what you're doing is working or not.

And so your choices are based on feeling.

And so you are able to feel your way through this problem and survive.

This is not a small advantage.

The ability to survive in unpredicted environments is an enormous adaptive advantage.

And so that, we believe, is what the function of feeling is.

That's why, in this narrow example, why respiratory control suddenly becomes conscious,

that it dramatically intrudes on consciousness suddenly, your need for oxygen.

And the purpose of this is to enable you to calibrate your choices, to change your mind about your policy,

your current policy, on the basis of whether it's working or not, which is exactly what feeling announces for you.

In the absence of feeling, you would behave randomly and one in a million will do the right thing.

And that one will survive and reproduce.

And the rest of you have had it.

So this enables us to make choices within our own lifetime.

We don't have to let natural selection do it.

Within our own lifetimes, we can adapt to unpredicted situations.

Of course, once you've done that, you then also can learn from the experience within your own lifetime.

And so the next time you find yourself in a burning building, you might, on the basis of learning from experience,

you might have somewhat less uncertainty about what to do.

So, again, let me just make sure that I'm getting across my main point, because this is the mechanism of feeling.

This is the function of feeling.

This is what feeling does.

This is why the organism must feel it.

Now, remember what Chalmers had said, that, you know, why doesn't all this information processing go on in the dark and so on?

That was built upon an earlier argument by Tom Nagel, who said, an organism has conscious mental states if and only if there is something that it's like to be that organism, something it's like for the organism.

Then he went on to say, if we acknowledge that a physical theory of mind must account for the subjective character of experience,

we must admit that no presently available conception gives us a clue about how this could be done.

So, this is what I'm trying to do in this talk.

I'm trying to give us a clue about how this could be done.

This is Nagel's way of formulating the hard problem.

Why and how is there something it's like to be an organism, something it's like for the organism?

My point is that that question only makes sense in relation to things like visual information processing,

which it does not have to be something it is like to see.

There does not have to be something it is like to discriminate red from blue.

You can do that unconsciously.

The cortex can do that automatically.

That kind of information processing can go on in the dark.

But would Chalmers even have asked, I mean, Nagel even have asked this question if we were talking about feeling?

I've just explained to you what the function of feeling is.

And it makes it kind of absurd to ask why and how is there something it is like to be an organism,

something it's like for the organism.

I hope that you can see what I mean.

Why and how there is something it is like to be an organism,

something it is like for the organism, has everything to do with feeling.

Feeling is what it is like to be an organism.

How you are doing as an organism.

How much longer are you going to remain in existence as an organism?

That's what feeling is like.

And this is why and how it exists.

So if we can get to what the physical mechanism,

going back to what Nagel was saying here.

He said, if we acknowledge that a physical theory of mind must account for this something it is likeness,

then he's saying we have to admit that we currently have no clue.

I'm saying it's because we've been looking in the wrong place.

I think if we look to feeling and we seek a physical theory of feeling,

this will account for the subjective character of experience and we will make some progress.

This will provide us some clue about how we might go about solving this heart problem.

Now, my next point.

Oh, so let me pause at that point.

Having said that affect is an extended form of homeostasis, that's my next claim.

So let's see if anybody wants to argue the toss or make a comment or ask a question.

Yeah.

So, Stephen, there's no need to raise hand.

After each point, we'll take any questions.

So go for it, Stephen.

And then Dave, if you have anything.

Thank you.

Yeah, thank you.

No, I'm really, really enjoying this.

I'm interested in this awareness that leads to the homeostatic sort of correction.

And if it's where the awareness is with the actual cells or organs struggling in some way to act or enact upon the surprisal,

be that in the environment that they're encountering at the different scales.

So this issue of actually acting or enacting and when that action and inaction can functionally correlate up the nested hierarchies,

then it's those actions or the inability to act in the way that cells would like that then is becoming the signal of what or where action is failing,

rather than necessarily a signal of that actually contains information inherently in it.

I wonder what your thoughts would be on that.

Yes, I agree.

If I'm understanding you correctly, I agree with the emphasis on action.

So I'll slightly run ahead of myself if I say this.

Obviously, a homeostatic deviation can also be construed as a prediction error.

In other words, what you're doing has not led to the outcome that was expected.

And there are two ways of correcting prediction errors.

You can either change your prediction or you can change what you're doing in order to bring about the prediction that you had originally.

In other words, when it comes to prediction error, you can either update the prior and have a posterior prediction,

or you've got to do something differently in order to confirm the prior prediction.

So that's where the emphasis is on action.

Now, why this is so important in relation to homeostasis is that you can't change your prediction about what your viable bounds are.

You know, if you expect to be between 36.5 and 37.5 degrees Celsius on the basis of acting in a certain way,

in other words, firing an autonomic reflex, and you then find yourself to be at 39 degrees Celsius,

you can't say, okay, my posterior prediction is that by doing this, I'll be 39 degrees Celsius.
Because if you do that, you're rapidly on the road to death.
So the prediction error has to be corrected on the basis of changing your action policy,
doing something different in order to confirm your prior prediction.
So the emphasis is very much on action when it comes to these organismic predictions,
these phenotypic, the viable states for your phenotype can't be changed.
So you can't just change your mind about what you expect will flow from your actions.
You have to change your actions in order to bring about the expected or preferred state of your phenotype.
Thanks.
Thanks.
Can I just add one piece to that?
I think that's really helpful.
And also in terms of like multiple scales going down to smaller scales.
So for instance, if I actually feel heat in my body and my body might be,
my cells are trying to act to find a better state.
That can be seen as oppressive and claustrophobic.
If I then read that as being me getting into an uncomfortable state,
it could also be me basking in the sun on a beach, you know,
in the same way that if I taste something that's very sour,
there's some sort of action at the cell or the organ level reacting to that sourness or that sweetness,
which could be like a nice confectionery suite for a child, you know,
like one of these gobstoppers, or it could be quite a problem.
So I think I like what you're saying with action.
I'm almost wondering whether that action piece is the inaction is the prediction error
in terms of how it can go up in terms of these nested hierarchies of physical scale.
Yes. So again, I must be careful not to run ahead of myself too much.
I mean, inevitably, you know, you must ask whatever question comes into your mind at any point.
But I'm aware that as I go down this list of arguments, I'm going to be addressing those points.
So I don't want to make all of those arguments in one go.
So I will just say that for now, that what you've just asked has, first of all,
a lot to do with the fact that we have multiple needs, that there's not only one need,
that they have to be balanced in relation to each other.
And also, very importantly, that we're talking about a predictive hierarchy,
that this is what you're speaking about with reference to scale,
that what applies at the sensory periphery,
and what applies at the core of the predictive hierarchy,
have different implications for the way that affect works.
So I'm just saying those very kind of vague and abstract things for now.

And I hope that the picture will become clearer as I proceed.
Thank you, Mark.
Yes, Dave, go for it.
Yeah, I don't know how to state this clearly.
This is some thinking that I got into from listening to a very recent interview you did
with a young man in another part of Cape Town.
We're talking about perception.
And I think there, and maybe this has already been done,
I just haven't run across the research,
that there's a larger framework in which these notions of what can be perceived,
what can be perceived consciously,
what is a sensory modality,
what we don't call a sense.
It seems that there are, on the one hand,
conditions, we've been calling those homeostatic conditions,
that you can't do anything about.
Your body either adjusts the blood pressure internally,
or it goes haywire, but you can't be aware of it.
So we don't think about this as something that we can consciously perceive,
and we don't call the mechanisms that are sensitive to blood pressure
and core temperature and other things as matters of sensation,
matters of perception, matters of consciousness.
On the other side, the other side of the mapping,
there are single object influences.
A mechanism that is sensitive only to blood pressure
presents another reason not to call this a sense.
An eye spot just tells certain kinds of worms,
there's light around here, or there isn't light around here.
No directionality, just it's daytime, or it isn't.
I'm under the mud, or I'm not under the mud.
Whereas even a little more information,
more kinds of information,
qualify this more as a sensory modality.
And in things like hearing and vision and smell,
we have this very rich many-to-many.
There's many things you can do about a burning smell,
and there are many kinds of odors out there.
Does this tie into anything you've been thinking about?

Yes.
So the first thing that you're touching on there
is the fact that homeostasis is only part of the story.
In fact, the example I gave earlier
of the person who finds themselves
in a carbon dioxide-filled room in a burning building,
they need to maintain their homeostatic blood gas balance.
But there's nothing that their autonomic nervous system
can do about that.
They therefore need to turn to action in the outside world.
So when I said that the person starts moving about,
upstairs and down,
and then feels,
is this working or is this not?
Is this good or is this bad?
That we call allostasis.
It's acting in the outside world
in order to return myself to my homeostatic balance.
So that's the first thing I just wanted to make clear.
I know it's implicit,
and I know I'm telling you something that you know very well,
but I just want to make that explicit
for our participants
that the importance of the outside world
for these internal bodily states,
the bridge from homeostasis to allostasis
is how we conceptualize that.
So then it moves to the next point,
which is that think about those kids
that I showed you earlier
who have no cortex
and all they're capable of feeling
is their affect.
They can't consciously think,
I feel like this about that.
In other words,
they don't know what the about that is.
They just feel things

and they can't include
within their consciousness of the feeling
what the thing in the outside world is
that contextualizes that feeling.
the context doesn't become conscious,
only the feeling itself.
To be able to extend the feeling onto the context,
to be able to say,
in effect,
I feel like this about that,
incorporates the context
within that terrain of uncertainty
where one's navigating,
one's palpating the uncertainty
in order to make choices,
not only on the basis of blind feeling,
but also on the basis of what kind of object
brings about this change in my feeling
and what kind of object brings about that.
For all of that to be incorporated
within the realm of consciousness,
I think is a further leap
and clearly another enormous adaptive advantage.
So the first leap is just to be able
to feel the consequences of your actions.
The second leap is to be able
to picture, as it were,
those actions within the sphere of consciousness
and in this way
to bring the context into consciousness
and extend the realm of choices enormously.
Thank you, Mark.
And I'll just chime in with a third question.
So you mentioned homeostasis
is only part of the story.
And in the chat, Brock asked,
what part of the story is hormesis?
I don't know that word.

So a small stress inducing,
like some kind of benefit
over the integrated time horizon,
like an exercise stress
that leads to improved strength.
You mentioned allostasis,
kind of an anticipatory movement
towards a set point.
But where does, for example,
induced stress, mild stress,
and recovery play into this extended homeostatic
or generalized homeostatic framework?
So, sorry, I didn't understand what you meant.
So, you know, I think, again,
when one gives an overview
of an argument like this,
one always oversimplifies,
one sort of has to simplify.
And, you know,
when I say that homeostasis,
that, you know,
you always have to confirm your prior prediction,
that's not entirely true.
There are also ways in which
the homeostatic range can be extended.
The homeostatic ranges,
I mean, like, for example,
you know, what I was saying about oxygen,
you can, if you're a diver,
you know, you can learn
how to hold your breath
and how to manage the stress
of being out of your viable blood gas range
in a way that a naive person like I cannot.
So there are mechanisms whereby
these things can,
in very narrow limits,
these things can be changed.

But I think the emphasis there
has to fall on the narrow limits.
Ultimately, there is an outer limit
to your viable range.
And, you know,
this is the driving mechanism
of the story that I'm talking about here.
Thank you.
Of course, so much more to say
and learn and add,
but please carry on.
Thank you.
Well, you can see highlighted
on the screen now my next point,
and it might seem like a small point.
It might seem like an obvious point
that complex organisms require
multiple homeostats.
And in fact,
I've more or less said this already,
but I want to make explicit,
and here is the same slide
that you saw a moment ago,
where I've just added the point
that I'm now making.
The importance of this point
is that we have multiple needs
which must be categorically distinguished
from one another.
In other words,
we can't have a continuous variable
called need.
That variable has to be factorized
across a number of different categories.
So let me be clear
what I'm talking about.
Imagine you have,
if we quantify on a continuous scale,

how much deviation there is
from where you need to be.
Let's say,

okay,
I've got six out of ten
of thirst,
and I've got four out of ten
of sleepiness,
then all I,
that means I've got
ten out of twenty
of total need,
and so all I need to do
is sleep.
I don't need to drink.
If I can generalize
from that little example,
the point that I'm trying to make
is that,
no,
that's not true.
You have to sleep,
and you have to drink,
and you have to eat,
and you have to defecate,
and you have to breathe.
You can't just summate all of these
and bring down the total number.
If you did that,
you would die.
So,
we have to recognize
that each one of these needs
has to be met
in their own right,
and that means
we need to know,

we can't just have
a total variable
called need.
We need to know
which need
are we talking about,
and this is why
they have to be treated
as categorical variables.
In other words,
this,
here we're talking about
how much water
I'm lacking,
here we're talking about
how much oxygen
I'm lacking,
here we're talking about
how much sleep
I'm lacking,
et cetera,
so that they can be,
so that the category of need
that the organism
finds itself in
can be addressed
appropriately.
The appropriate category
needs to be addressed.
And now,
I know what I'm saying
sounds absolutely obvious,
but why it's important
is that
it's because of this,
because of the point
I've just been making,
that hunger feels different

from thirst,
and thirst feels different
from suffocation alarm,
and suffocation alarm
feels different
from sleepiness,
and sleepiness feels
different from pain,
and so on,
because each one
of these feelings,
the quality of the feeling,
tells us which need
is at issue.
And categorical variables
are distinguished
qualitatively.
So we're not only
talking about valence,
goodness or badness,
and arousal,
how much or how little.
We're talking also
about qualitative categories.
And I think that's important
when we come back
to the whole point
of what I'm talking about,
namely qualia,
namely,
the qualitative stuff
of what it is like
to do anything,
be anything.
Those qualitative distinctions,
it's of the essence
of consciousness,
that it is qualitative,

it has qualitative
differentiations.
It's not just some
total quantitative,
some total continuous
matter of need.
It's a matter of different
categories of need,
which therefore
are different qualities
of need.
And I think that's important
in terms of understanding
what an affect is.
An affect is a state
of the organism
registered by the organism.
In other words,
it is intrinsically subjective.
It has intrinsic value,
valence,
goodness and badness,
which has existential consequences
for the organism.
And in addition to that,
it has quality.
It is inherent
in the nature of affects,
that they are qualitatively
distinctive from each other.
And I hope that
this very simple
functional mechanistic reasoning
that I've given you
as to why that needs
to be the case,
why that must be the case,
helps to make clear

why affects take the form
that they do.
Remember,
what we are doing here
is addressing
Nagel's question
about,
you know,
why is there something
it is like
to be an organism?
And how does it come about
that there is something
it is like
to be an organism
for an organism?
That something
it is likeness
is not just
a valence,
goodness and badness,
but also a qualitatively
differentiated state,
subjective state,
of the organism.
So I think that
by reducing
the mechanism
of affect
to these essential features,
we begin to see
why it takes
the form that it does,
why it is
a necessarily
conscious something
in the sense
that I'm,

the rudimentary sense
that I'm defining
consciousness.
So that's why
this point is important.
That's why
I wanted to make
the point clearly
that complex organisms
require multiple
categorical homeostics,
stats.
They need to be treated
as categorical variables.
They need to,
therefore,
be,
they need to be qualified.
Each of these
different factors
of need,
each of these
different categories
of need
have to be
qualitatively
differentiated
from each other.
And this is,
I think,
the ground zero
of where
qualia come from.
Now,
I'll go on
to my next point.
which is that
these different

categories of need
must be prioritized.
Again,
I think this is,
in a way,
an obvious point.
But I want to just
dwell on it
for a moment
to draw out
some of the
implications of this.
I said earlier
that our
autonomic
homeostats
run,
well,
autonomically
most of the time.
In other words,
there are
automatic,
automatized predictions
which we call
reflexes
that are,
when the
organism moves out
of its viable bounds,
then it fires
these reflexes
which bring it
back into its
viable bounds.
And then I told you
that not all
of our needs

can be dealt
with that way.
Some of them,
as we move
outside of the
range
of what
the
autonomic reflex
can achieve,
like,
for example,
when it comes
to thermoregulation,
you start
to perspire
if you get
too hot.
This is a reflex.
You start
to breathe
more shallowly
and rapidly.
Panting,
in other words,
that's a reflex.
These are
autonomic responses
to overheating.
Then they reach
a certain limit
that they
haven't cooled
you down
adequately.
So then you've
got to do
something.

You've got to do
something like
leave the kitchen.
And that is
what I've just
said applies
to one
of your
multiple
homeostats.
In other words,
one of them
now requires
allostatic action.
You now need
to do something
in the outside
world.
And clearly
you can't do
something in the
outside world
in relation to
all your needs
simultaneously.
So all the
time you're
sliding,
you know,
your hydration,
your water
in relation
to salt
content,
of your body
sliding all
the time.
It doesn't

mean that you're
thirsty all
the time.
You're burning
up the sugars
in your adipose
tissues all
the time.
It doesn't
mean you're
hungry all
the time.
so the question
is, you know,
what gets
prioritized?
And there
must be some
prioritization
because you
can't do
everything at
once.
When it comes
to voluntary
activity in the
outside world,
you have to
prioritize.
There's an action
bottleneck.
And the point
I'm making
here is,
first of all,
just simply
that, that you
do have to

prioritize.
And secondly,
that what is
prioritized is
what becomes
conscious.
In other
words, as I
gave in the
example of
suffocation
alarm, that
business carries
on unconsciously
until it becomes
prioritized.
At that point,
it forcibly
intrudes on
consciousness.
My priority
now is I
need oxygen.
And that's
what you feel.
The other
needs, this is
the other
point I'm
making, the
other needs
don't disappear
at that point.
They continue
to be
regulated, but
they are
regulated

automatically.
So the
prioritization
of a need
brings that
need into
the realm
of palpating
uncertainty.
In other
words, into
the realm
of feeling
your way
through the
problem.
And this, I
think, is an
important part
of how
feeling works.
So let me
hear, remember
I said I was
going to talk
about the
periaqueductal
gray.
And I was
going to tell
you how this
plays as
important, if
not more
important,
role in
the basic
machinery of

consciousness of the upper brain stem than the reticular activating system.

The periaqueductal gray, which is just a 14-millimeter long structure, columnar structure around the central canal of the midbrain, all of our homeostatic mechanisms, in other words, all of the multiple homeostats, and we have many, all of the multiple homeostats, all of them send their residual error signal to the periaqueductal gray.

The periaqueductal gray is like a final common pathway of all

of these
homeostatic error
signals.
And it seems
that this is
where the
prioritization
must be going
on.
The determining
which of the
current error
signals is the
most salient
then gives
rise to, and
this is an
argument I must
again make
clear, this is
not my own
novel insight.
This was
beautifully
argued on
comparative
anatomical
grounds by
Bjorn
Merker in
a brilliant
paper in
Behavioral and
Brain Sciences.
So he
recognized that
the periaqueductal
gray, together

with the
superior
colliculi that I
showed you
earlier,
and the
midbrain
locomotor
region, they
form what he
called a
midbrain
selection
triangle.
In other
words, this
is where the
need that is
to be
prioritized is
selected, and
this gives
rise to a
feeling, the
feeling being,
as in the
example that I
gave with
suffocation
alarm, the
need that is
prioritized, the
need that is
currently most
salient, the
need that is
going to now
color your

consciousness,
qualify your
consciousness in
terms of its
most rudimentary
property, in
other words, the
feeling, the
affect, what
state, what
organismic state
am I in, I'm
in a state of
respiratory distress,
I'm in a state of
suffocation alarm,
that's what I
feel, why?
It's because
this is where
choices need to
be made, this
is where the
creature needs
to feel, the
person in that
example needs to
feel their way
through the
problem.
The other
needs remain, but
the other needs,
the non-prioritized
needs are not
raised to the
level of feeling
and I hope it's

clear why.
And we need to
understand
mechanistically how
does all of this
work and that
leads to my next
point.
So before I go to
that next point,
let me just pause
again in case there
are any comments or
questions about this
prioritization
function performed
by the periaqueductal
gray, which is
what determines
what affective
state you are
going to be in
from one moment
to the next.
In other words,
which need is
going to qualify
your affective
state.
Thank you,
Mark.
Dave, Ben,
Stephen?
Yeah, I just
want to throw
out there, in
Sylvan Tompkins'
later work, that

is after 1970,
he came to
elevate
reset
to the status
of a
fundamental
emotion.
And I think
it's very relevant
to this, the
notion of
suddenly shifting
from one
conscious
emotion or
one conscious
activity or
world attitude
to another.
I believe
Yak Pongsep
drew pretty
heavily on
Tompkins, and
the reason that
he doesn't cite
him in a lot
of detail is
that he accepted
so much of
it, kind of
like what you
said about,
I agree almost
totally with,
oh, the
Britisher whose

father's from
India, I'm
sorry, I'm
blind for
knowing his
name, you
had a dialogue
with him just a
few months ago.
Am I
completely wrong
about that?
Neil Seth are
you talking
about?
Yes, thank
you.
Am I
completely
wrong?
Did
Tompkins
simply agree?
No, I agree
with that
completely.
Tompkins, the
reason that
Pongsep doesn't
cite him, or
barely ever, he
cites him
usually in the
beginning of a
kind of a
general
discussion, you
know, just to

say this is the
tradition I belong
to, and then
carries on
building on his
shoulders.
I agree with
that very
much.
Thank you.
Stephen?
Yeah, I
liked when you
talked about
palpating
uncertainty and
the salience
around that, so
that's a nice
term, and I
think that
relates to
what is going
on, so to
speak.
So what about
what's going
on, and then
now what
happens next?
And I'm
wondering,
though, as
well, if the
idea of
where, which
I think is
often left

out, there's
a lot of
categorization of
what things
are, but
where is it
happening?
That makes a
difference in
terms of how
significant it
might be in
terms of
prioritization.
So I was
wondering how
much the
risk of the
awareness of
some sort
of disequilibrium
happens, could
be featured in
this, and
maybe that's
where the
physicalization
and the
spatialization
becomes much
more significant
than when you're
just dealing
with the
brain.
Yes, so
well, all of
these homeostatic

mechanisms, of
course, the
physiological
processes that
they regulate are
widely distributed,
but the
control centers
of these
homeostats are
for the most
part in the
brain stem
broadly defined,
and I say
broadly defined
because there's
some that are
in the medulla
oblongata and
pons, but
there's some
that are in
the midbrain,
there's some
that are
diencephalic.
For example,
I mean, to
mention the
most outstanding
example, the
hypothalamus is
full of
homeostatic
control centers.
But, you
know, so

hypothalamus,
circumventricular
organs, the
parabrachial
complex, the
area postrema,
the nucleus
solitaris, these
are all
homeostatic
control centers,
but of
fundamental
importance, and
I think it's
not sufficiently
recognized,
although Marcus
certainly made
much of it,
that all of
these nuclei
that I have
just enumerated,
they in turn
project to the
periaqueductal
gray.
The periaqueductal
gray is, in my
view, a
metahomeostat.
It's sort of
the control
center of the
control centers,
and the
center of the

prioritization
function.
And by being
able to
physically locate
these mechanisms,
bearing in mind
where I
started, I
started my
answer to your
question by
pointing out that
ultimately these
homeostats regulate
physiological
processes which
cannot be
localized.
localized.
They are
distributed
processes,
par excellence,
but the
center, the
control center,
can be
localized, and
those control
centers are,
there are
numerous ones,
and they in
turn, they all
send the
residual error
signal to

periaqueductal
gray.
That is really
important.
Now what that
enables us then
to do is to
test models like
the one that I'm
describing to you.
What happens
when one or
another of these
individual
homeostatic
control centers
is lesioned?
What happens
when the
periaqueductal
gray as a
whole is
lesioned?
What happens,
I might as well
just insert that
here, is you
get a
persistent
vegetative
state.
In other
words, you
get that
condition that
Magoon and
Maruzzi led
us to

believe
is a
theoretical
possibility.
In other
words, blank
wakefulness.
Remember I
said earlier
in my
summary of
what I
said last
time I
spoke, that
the idea
that the
reticular
activating
system provides
merely a
quantitative
level of
consciousness
and not
any quality
or content,
that
fictional
state of
affairs, of
blank
wakefulness,
the thing
that the
reticular
activating
system was

supposed to
be producing,
which as I've
showed you
is not the
case because
all of those
different lines
of evidence
that I
summarized show
that what the
reticular
activating
system is
producing is
anything but
without quality
and without
content, that
this is in
fact the
foundational
form of the
qualities and
contents of
consciousness,
namely the
different
affects.
if you
lesion the
periaqueductal
gray, then
you get that
fictional state,
that artificial
state of

blank
wakefulness.
In other
words, these
patients show
non-responsive
wakefulness.
They still have
the autonomic
sleep-waking
cycle.
In other
words, they
wake up in
the morning and
go to sleep
at night like
those hydran and
cephalic kids that
I showed you.
But unlike
those kids, they
show no
emotional, no
affective, no
response to
their situation
and no
intentionality
either.
And so, you
know, that's
what you would
expect would
occur if the
periaqueductal
gray is where
the affects are

actually being
generated.
Because remember,
that's what I'm
saying.
I'm saying that
the prioritization
of a homeostatic
error signal is
the feeling of
that signal.
It is the
rendering conscious
of that signal.
And that is
what we mean by
feeling something.
It means now
this need is not
just a need.
Now this need
is a drive.
Now this need
is driving my
voluntary behavior.
And that is why
it becomes conscious
because as I said
to you earlier,
voluntary behavior
can be defined as
the making of
choices as
opposed to the
executing of
automatized
predictions.
Just to add

two notes on
that, when
hearing Stephen's
questions about
spatial localization
and how that
can refer to
brain localization
as well as
like in the
peripersonal space,
it made me think
about how
awareness,
a-wareness,
it could be
without awareness,
awareness could
be spatially
dislocated,
or awareness
might be a
highly spatialized
percept.
And then the
other note that
I wanted to
make was as
a researcher of
distributed
physiology in
the eusocial
insect colony,
this extended
homeostatic
perspective leads
to many
interesting

connections when
we think about
extended social
homeostasis as
well.
But just two
notes and we
can carry on.
Very good
points.
So you're
saying we can
carry on,
meaning I can
carry on?
Whichever way
you'd like to.
Okay, great.
So I'm now
going to move
to my next
claim, which
is, I see
we have half
an hour left,
so I mustn't,
it's really
quite amazing,
you know,
because you
are so generous
with giving
your speakers
two hours,
you lull us
into a full
sense of security
and thinking,

well, I can
just elaborate
as much as I
want to,
there's plenty
of time.
And now we're
down to our
last half hour
and I've still
got three
points to go.
It's like no
one goes there
anymore, it's
too busy.
Yeah.
Well, I'm
again, grateful
for the time
that you're
giving me.
So let me
move to this
point.
And this has
everything to do
with the
relationship
between affective
consciousness
and cognitive
consciousness.
I spoke earlier
about, I feel
like this,
about that.
In other words,

the incorporating
of the
cognitive domain
within the
sphere of
feeling.
And so this
is the point
that I'm
making now.
This is a
drawing that,
a diagram that
comes from a
paper that I
wrote with
Carl Fristen.
And here,
the important
equation is
the third
one.
And here,
this diagram
basically is
just trying to
spell out in
a visual form
what these
equations are
saying.
So when I
said earlier
that we
have two
ways of
reducing
prediction

error,
when a
prior prediction
does not
lead to the
sensory state
that's
expected,
then we
can either
change our
perception,
in other
words, we
can change
the prediction,
or we can
change our
action.
We can do
something
differently in
order to bring
about the
prior prediction.
So that's what
these two
equations describe,
that we have
a generative
model which
generates
predictions as
to what
sensory state
will flow
from our
actions.

And so
here, there's
an action in
the external
world which
is predicted
to bring
about a
certain sensory
state, and
to the extent
that it does
not bring
about that
sensory state,
in other
words, the
difference here
is the
prediction
error.
And of
course, the
prediction error
is then used
to update
the generative
model in
order to
give rise
to better
predictions, in
order to
better maintain
your expected
sensory states.
I need to
emphasize here,

because this
was not the
case in the
early days of
the predictive
processing
paradigm that
Carl Fristen
and his
colleagues
unleashed upon
the world,
sensory states
do not
necessarily mean
extra-oceptive
sensory states.
So in the
examples that
I have been
giving in this
talk so far,
this would be
the equivalent
of, for
example, core
body temperature
or blood
oxygen level,
et cetera.
The most
important sensory
states for the
organism are
its viable
states in terms
of its
homeostatic

expectations.
actions.
So please
remember that
when I speak
about actions
in the world,
the world that
we're talking
about here
can be the
visceral body
and it can be
the external
world.
The body,
the Markov
blanket,
the body is
as much
external to
the blanket
blanket as
the external
world is
external to
the blanket
of the
nervous system.
I think we've
got some,
we're in
somebody's
kitchen here,
which you
must put,
you must
put your

mic,
whoever is
doing their
dishes there.
So,
what this
equation here,
which is
written in
words here,
the rate of
change of
precision,
precision,
which is
omega in
the equation,
the rate of
change of
precision over
time depends
on how much
free energy
changes when
you change
precision.
This means
that precision
will look as
if it's trying
to minimize
free energy.
The rate of
this free
energy
minimization
process is
the difference

between the
inverse
precision and
the sum of
squared prediction
errors.
So that's
this equation
here put into
words.
And it's
foregrounding
the central
role that
precision plays
in minimizing
prediction error.
So I
want to be
clear that
this is
precision
modulation
is in
physiological
terms the
modulation of
post-synaptic
gain.
So the
message passing
going on
here,
between
error
signals
and
prediction

signals
is
modulated
by
posts.
It's the
reticular
activating
system's
modulation of
that message
passing.
So it's the
increasing or
decreasing of
the gain on
the error
signals.
That's the
role of
precision
modulation,
which is just
the same
thing as to
say that is
the role of
the reticular
activating
system.
It is
modulating
the gain
in the
message passing
to speak
physiologically.
To speak

computationally,
it is a
matter of
increasing
or decreasing
precision values
attached to
the predictions
over the
errors.
error
is bad.
If
things are
turning out
as expected,
that's good.
If
uncertainty
prevails,
that's bad.
So increasing
confidence in
a prediction
is good.
In other
words,
increasing
precision in
a prediction
is good.
Increasing
confidence in
an error signal is bad. In other words, the more uncertainty, the more you become clear that things
are not turning out as expected, to that extent, of course, your confidence in your current policy
is reduced. In other words, the precision in your current policy is reduced, and that just is bad.
So the goodness and badness, the pleasure and unpleasure function of precision modulation that I described to you earlier,

it has this enormously important contribution to make to the whole of this mechanism by determining

the influence of the error signal over the predictive model.

So to the extent that precision is reduced in the error signal, and thereby confidence is maintained in the policy,

to that extent, the error signal will or will not have influence over the parameters of the predictive model.

And so this is trying to illustrate the crucial role that EPIC plays in this whole predictive mechanism.

So this is, as it were, the role of the reticular activating system, of the modulation of the message passing

that goes on. These are, as it were, the synaptic transmission mechanisms,

and these are post-synaptic modulatory mechanisms. And so I'm just wanting to link these formalisms

to the affective role that the reticular activating system plays.

So this, so just to go back to the statement here, the mechanism of perceptual and cognitive consciousness

is precision modulation of allostatic prediction errors. In other words, it is the modulation of confidence

in a current policy. In other words, the perceived consequences of a current...

So as you, as the, as the, a particular need is prioritized, that need is felt.

This generates a category of predictive policies. In other words, this is what I will,

this is what I do in this situation, in this context that I find myself in.

This is what I expect the consequences will be. In other words, these are my expected...

And there's an expected precision attaching to the error signals, of course.

All the other domains of need, remember what I was saying earlier here about the needs must be prioritized.

The other domains have monotonous precisions.

So the crucial mechanism in terms of voluntary action is that, to use the phrase that one of you said you liked earlier,

is the palpating of the uncertainty. In other words, the palpating of the precision

and the adjustment of the confidence in the current perceptual and cognitive, in other words, the allostatic aspect of what I must do about this need state.

Because there's a state of uncertainty that's been prioritized there, there's a changing of one's mind on the fly.

This is what this mechanism makes possible.

So it is the affects are, as it were, a drive for or a demand for predictive work, for mental work.

And this, the modulation, the palpating of the confidence in the policy over the error signals is the predictive work so demanded.

So in other words, the work of updating one's policy, of changing one's mind, it is all of it underwritten by the affective demand.

So the affective demand for predictive work is what gives rise to the predictive work itself, which is the palpating of uncertainties in the current policy and the sensory states that it gives rise to.

And so this is changing your mind. In other words, in other words, voluntary action.

In other words, the capacity for choice. This is the crucial role that precision modulation plays.

In that process.

So that's, oh, okay, good.

So I think, let me just pause for a moment to see if there are any questions about that, because I wanted to illustrate it with a case, which I think is going to pretty much take us to the end of our time.

So before I launch into this last case, let's see if anybody has any questions or comments about that mechanism.

I think we'll go to the case and then there's always so much to unpack with defining the variables and understanding what the edges mean and so on. But I think the case is a good way to close this. Okay. So this is a patient of mine, Mr. S. And he had a meningioma here at the base of his frontal lobes. It was an olfactory sheath meningioma.

And it pushed on his optic nerves. And as a result of that, his vision was impaired. And this is how the tumor came to attention. And it was successfully surgically resected. Now, because of its location, in relation, in fact, to the optic nerves, there was some nervousness on the part of the surgeon to remove it in its entirety. In fact, he felt it was not possible to remove it in its entirety.

So he left a little nub of the tumor. And this re-grew. And so the patient again noticed these visual difficulties, returned to the surgeon, and the operation had to be repeated.

Because of scar tissue, the second operation is always trickier than the first. And unfortunately, in the second operation, there was a bleed. And that bleed was into the basal forebrain nuclei. Small bleed, but the basal forebrain nuclei are crucial. These are the upper end of the, of the, remember I was saying earlier that these different parts of the ascending arousal mechanisms of the, of the brain, that they, that they are the source nuclei for these different neuromodulators. And I spoke of dopamine and noradrenaline and serotonin earlier. Well, the, these nuclei are the source nuclei or, or, or, or, there are other source nuclei for acetylcholine here too. But these are very important source nuclei for acetylcholine. Now remember, all of these, what I was saying earlier about these, these neuromodulatory systems is that they are modulating postsynaptic gain, which is just the same thing physiologically as, to say computationally, they're modulating precision. That's what they do. They up and down regulate the precision in the message passing. And this, as I said earlier, it, it dictates, you know, which, which messages are going to, are going to be selected and, and which not. So it's a, it plays a crucially important role, does precision modulation in cognition. And, and I'm now wanting to show you this relationship, the relationship between the affective mechanisms that I'm talking about, that is the affective functions that are performed centrally by periaqueductal gray, and then how this, it gives rise to the modulation through these arousal systems of these different neurotransmitter systems. And I'm going to now show you how this worked in this case. Acetylcholine modulates the confidence in error signals. And so, let me just quickly tell you a little bit more about this patient. He was, he was 56 years old, at the time of the second operation. And as a result of the damage to the basal forebrain nuclei, from the second operation, he woke up from the surgery with a condition called confabulatory amnesia. So although the tumor was resected successfully, and the visual problem was, was, was corrected, he now had this devastating new condition called confabulatory amnesia. Confabulatory amnesia, the patient is, is, it's not only that they are amnesic, in other words, that they're unable to remember, particularly recent events, but there's also quite a long retrograde extension too. In other words, there's quite an impairment of the retrograde memory too. Not only do they have this, but they also are not aware of their, of their memory problems. And so when they draw up, when they attempt to retrieve a memory, the memory that they retrieve is not the correct one. And they don't realize it's not the correct one. And so they have what appear to be false memories. These are the confabulations, they frequently are related in, in, in some semantic sense with the target memory with the memory that they're looking for, that there's some semantic relationship between the memory they find and the one they're looking for, but they can be grossly mis, misplaced in space and time. And so these are what we call confabulations. And the patient is not sufficiently critical of these misrememberings, as a result of the, the damage to these neuromodulatory mechanisms. So this patient, Mr. S, just to give you one example, I mean, an, an, an, an, an

extreme example. He, he had his operation in Johannesburg, which is in South Africa, which is where I hail from. But at that, at the time, I was living and working in London. And so the surgeon who I knew well, sent the patient to see me to consult me in London.

And because I was doing work with this, with this condition, confabulatory amnesia. So the patient arrives in London on a Friday, and comes to me on the Monday, and, and has no idea that he's in London, because of course, he doesn't remember the journey, that he can't remember anything from one minute to the next. And so I say something about well, you know, the, the surgeon, whose name was Mr.

Miller, the surgeon referred you to me because of this memory difficulty that you're having. That's why he sent you to London. And he said London. What do you mean London? And I said, Yes, you know, you're in London, you don't realise it because you don't remember the journey. You know, that's the whole point. This is the kind of problem that you have in London.

He said London. He denies it. He says he's not in London. Now it's certain that so that's the confabulatory aspect that these patients that he is memory, he's in Johannesburg. And so he believes he is in Johannesburg. So I point out to him, it was winter.

he says he's not in London. Now, so that's the confabulatory aspect that these patients, in his memory, he's in Johannesburg. And so he believes he is in Johannesburg. So I point out to him, it was winter. And it was it was snowing outside, I point out to him, the the wintry conditions outside, you never, by the way, have snow in Johannesburg. So I say, look out the window, he looks out, he's absolutely shocked. But then retorts, no, I know I'm in Johannesburg, just because you're eating pizza doesn't mean you're in Italy. That's what he says to me. So in other words, you know, you don't have to, you know, you mustn't overrate the evidence of your senses. And so that's just an extreme example of what I mean by how these patients, how, how, how them, their amnesia, it's not just a lack of memory, it's also an excessive confidence in the, in the incorrect memories that they draw up. So that's the that's the background. And I saw this man, the whole point of him being referred to me is because I treat such patients. And so I then saw him six days a week. At the same time in my outpatient clinic, at the Royal London Hospital, same time, same place, he came with his wife to the waiting room, I would then go collect him, take him up to my consulting room, spend an hour with him, take him back down to the waiting room. And then I would talk to his wife, because he was so full of confabulations, I needed to verify things in order to get some sense of what was going on. And there were certain themes. And this is an interesting thing about these patients, these confabulatory patients, there's certain themes that returned again and again, that the patient was an electronic engineer, but but he, in reality, he was, but he thought the reason he was coming to me frequently, he thought he was coming to me to, to, to, because I was consulting him about an electronic problem. And, or otherwise, he thought that the two of us were electronic engineers together, working on some electronic problem. Also frequently, he thought that he and I were in some sporting team together that we played rugby together, or, or we, we were in this rowing team together. Now his wife told me he had played rugby at university, a good 30 years or more before. And likewise, he had, he had been a keen rower. But this too, was at university more than 30 years before. And so these are, again, good examples of confabulations, you know, the patient, you know, the patient, mislocating in space and time, a memory that, that he draws up now, and he has too much confidence in that, he's too, too, too readily accepts the veracity of the, of the products of his own memory search. So that's the, that's the background. Now, on this particular day that I want to give you a little snippet of a session from, he, on this day, when I came to the waiting room, he touched the scar on his head, the craniotomy scar on his head. And he said, Hi, doc, to me. So this was progress, you know, for the first time, he was associating me with medicine. And he was associating me with the surgical scar on his head. And so I thought this was great progress. And so when we got into my consulting room, I said to him, you touched your head when we met in the waiting room. And he said, I think the problem is that a cartridge is missing. We must, we just need the specs. What was it, a C49? Should we order it? So I said to him, what does a C49 cartridge do? He says, memory, it's a memory cartridge, a memory implant. Oh, sorry, I should have mentioned this. His wife told me that he had had dental implants. He'd had serious problems with his teeth. And these had been for many years, but these problems had finally been successfully treated by implants, teeth implanted into the into the jaws. So when he said that that C49 cartridge is a memory cartridge, and that it's a memory implant, it brought that operation to the dental operation to my mind. He said, But I never really understood it. In fact, I haven't used it for a good five or six months now. It seems we don't really need it. It was all chopped away by a doctor. What's his name? Dr. Solmes, I think. But it seems I don't really need it. The implants work fine. So I said to him, you're aware that something's wrong with your memory, but and he interrupts me. And he says, Yeah, it's not working 100%, but we don't really need it. It was just missing a few beats. The analysis showed there was some C or CO9 missing. Denise brought me here to see a doctor. What's his name again, Dr. Solmes or something. And he did one of those heart transplant things. So this is referring to another operation that he'd had, which is clearly also being referred to here, which was that he had a cardiac arrhythmia. And so he had a pacemaker fitted, a cardiac pacemaker. So he says, So he did one of those heart transplant things. And now it's working fine. Never misses a beat. So I said to him, which is what I actually thought, I said, You're aware that something's missing, some memories are missing. And of course, that's worrying. You hope I can fix it, just like those other doctors fixed the problems with your teeth and your heart. But you want it so much that you're having difficulty

accepting that it's not fixed already. So he goes on and says, Oh, I see. Yes, it's not working 100%. And he touches his head. He says, I got knocked on the head, went off the field for a few minutes. But it's fine now. I suppose I shouldn't go back on. But you know me, I don't like going down. So I asked Tim Noakes, Tim Noakes, by the way, is a sports physician in South Africa. He says,
So I asked Tim Noakes, because I've got the insurance, you know, so why not use it? Why not go to the best? And he said, Fine, play on. So that's the end of the snippet that I wanted to show you. But by the way, before I show you about that, so so this is a case in which due to damage to one of the precision modulating nuclei, or sets of nuclei, the in this man's case, the basal four brain nuclei, due to the damage to the acetylcholine modulating mechanisms that I that I showed you earlier, he has too much confidence in his predictions, and too little confidence in the world. He does not up the game on the area. So he looks, I say to him, his predictive model tells him he's in Johannesburg. I say to him, No, you're in London. Look out the window, he looks out the window, he sees snow. You know, clearly not something that could possibly be associated with Johannesburg. But he sticks with his prediction. He says, No, I know I'm in Joburg. Just because you're eating pizza, that doesn't mean you're in Italy. And likewise, in the case of the these memory processes that I showed you in that session, each time that he starts to feel the unpleasure of the mounting error signals, the mounting that the things are so just think about it, he has, he touches his head is he's on the brink of being aware that he's he's had a brain operation, that the brain operation has resulted in loss of memory, I hope you can see that him saying that I he says, we just need to order the specs for this module that's missing, what does the module do it does memory, you know, so touching his head, speaking about operations. So he's on the he's on the brink of recognizing that things were not as he expected, that things are, in fact, quite different. My point being that this evokes feelings, that this is bad. This is a this is a panic inducing situation. And because this man has damage to these precision modulation mechanisms that we've been talking about, that he what he does is he simply upregulates the or maintains his confidence in his prediction, and in this way, maintains his his emotional equanimity, rather than allowing the unpleasant affect to dominate and the unpleasant, unpleasant affect to update to doing to to upregulate the error signal, and and which would which would normally update the predictive model. So I'm hoping that in this case, you can see something of the role of affect, the role that the feelings involved in this man's case, in relation to his predictions, and the prediction errors, in other words, the cognitive business of what he perceives, and what he believes, and how what he perceives, changes what he believes, and the role of feeling in all of this, I thought this was a succinct case to be able to illustrate, illustrate all of that.
The because it's only one case, I thought I should just show you some of this. These are several papers that we've written on showing how these mechanisms work in confabulation. We it's not just this one case, we were able to show that affect regulation, that that the confabulation, that confabulation has a wishful quality, in other words, it has a down regulating of error signals quality and an up regulating of predictions quality. And we were we were also able to show that by analysis of transcripts of cases like this, we studied many cases like this, how the affect actually improves with each confabulation. So

there's an increasing negative affect followed by confabulation, followed by an improvement in the
affect. So I see I'm right at the end of my time. I just want to, I don't have time to go into this
point. I'm sorry. This is this is not such a fundamental point. It's just a slightly different
way that I see the predictive hierarchy. If we if we take seriously that what we're talking about here,
fundamentally, we're talking about homeostatic systems, and that the most important predictions
have to do with maintaining your phenotypic prior preference distribution, that this has implications
for how we conceptualize the predictive hierarchy. But as I say, we don't have time to go into that.
And then the last thing is just to say that these models that we that I've together with my colleagues
derived from the study of of the neuroscientific evidence
on the basis of this well known statement of of Richard Feynman's, what I can't create, I do not understand.
We we are trying to instantiate these mechanisms that are described to you earlier in an artificial consciousness.
I'm working with a group of really great guys,
scientists and computer scientists and applied mathematicians. And we presented our preliminary findings
a few months ago at Carl Fristen's theoretical neurobiology meetings. And so watch the space. We are
we are we are we are we are we are saying that if we have identified the mechanism, the causal mechanism, where
whereby affects are generated, what they actually do in relation to a self organizing system,
then if this really is how affect is is generated, if this really is how it is caused,
then we should be able to instantiate it in a in an artificial system. And that's what we're what we're
trying to do at the moment. I'm sorry to have rushed at the end. But if you'll indulge me, I will wait for
if there are any questions, comments. I just wanted to show you my so this is I'm saying these questions can
be that they look quite different. If we look at it through the lens that I've tried to encourage you to
look at it with me through, in other words, through the lens of affect, these questions, why and how is
there something it's like to be an organism, something it's like for the organism, and questions like, why
is the performance of these functions accompanied by experience? This is Chalmers' question. Why doesn't
all this information processing go on in the dark, free of any inner feel? I'm saying that these questions,
questions, I've tried to provide you a clue as to how we might be able to address them differently.
Oh, what happened there? I've lost my
Oh, my good. Okay. Ah, there I am. Yes. How we might be able to address them differently, but
get them through this lens. Sorry to have gone on my word. I can't believe I've spoken for two hours
again. But if you'll, if you will allow me, I'll be happy to wait for any further comments and questions
before we end. As we have actually a whole document of questions and many questions in the chat, I think
it's a perfect place to end the live session. And whenever the affordance presents itself, we're always
happy to have a dot three, the numbers keep counting, and we'll keep hosting the sessions because this was
very fascinating. And I hope people got a lot out of it and very tantalizing.
Thank you so much for inviting me and apologies for going on for so long.
I really enjoyed what little interchange we had over these four hours. I really enjoyed. Thank you so much.

Maybe one nice future event could be picking up on those last two, which are
extremely interesting and important. And then starting with those two with the dot one and the
dot two as prerequisite, and then sharing more about those two when the time is right and having
that as more of a full discussion. Very good. Thank you, Mark. Thank you, Steve. Thank you, Dave.
Thanks, Stephen. Thanks, Dave. Thanks, Daniel. Peace. Take care. Bye.
All right. Thanks, everybody for watching. Hope you enjoyed that. See you later.